

AI in Medical Education: A Workshop by Students, for Students

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Abstract

Medical students must gain competencies in AI literacy in order to engage with a rapidly evolving clinical landscape. However, limited formal training opportunities currently exist. This paper describes innovative, peer-led workshops as an effective tool to bridge these gaps and facilitate meaningful gains in knowledge and confidence.

Keywords

Artificial intelligence; medical education; student-led; workshop; peer teaching

Article

Artificial Intelligence (AI) is rapidly transforming healthcare, from clinical decision support tools to diagnostic algorithms. [1] As such, there is a need for medical students to develop the competencies to engage with this evolving landscape. [2] Medical students have expressed interest in an updated curriculum that aims to increase familiarity and comfort with AI while protecting important professional values. [1] Current medical programs that have integrated AI education have been well received by medical learners. [2]

Despite this need, many medical students report having a lack of formal training in AI. [1] This educational gap limits the ability of students to navigate AI-integrated clinical settings and to critically evaluate the potential risks, benefits, and uses of these tools. Without proper training, future physicians risk implementing AI tools without understanding their implications on patient care and safety. There is therefore a need for widespread, formal curriculum to support medical trainee knowledge acquisition in this space. [2] However, the process of formal curriculum integration can be lengthy, which may leave learners at risk of completing their training without access to this knowledge.

To bridge this gap, we organized a series of four, one-hour long, interprofessional, student-led workshops within the Faculty of Health Sciences at McMaster University in 2024 and 2025. A total of 109 learners participated, representing medicine, nursing and other allied health programs. Participants were asked to complete pre- and post-surveys to improve the quality of the workshop for subsequent iterations.

The workshops were intended to serve as a primer for students with limited prior experience in AI. We first provided participants with an overview of AI; machine learning, including supervised and unsupervised learning; deep learning, and generative AI. We then provided examples of machine learning and generative AI in healthcare. We discussed how AI algorithms are developed and integrated, specifically through feature selection, training, and validation. This provided the foundation for a discussion on the pitfalls of AI, including flawed or biased data; algorithmic flaws, including overfitting and incorrect assumptions; black boxes; and hallucinations. We briefly reviewed a framework for the appraisal of AI, the APPRAISE-AI tool, [3] and finally had participants collaborate in small groups to review case studies and critically appraise AI and its implementation themselves in simulated scenarios.

81 of 109 attendees (74% response rate) completed the pre- and post- surveys. Participants were able to identify, on average, double the number of clinical applications compared to before. Moreover, 90.1% of participants could apply the term “supervised learning” in a multiple choice question after the workshop, compared to 24.7% before ($p<.001$). Moreover, participants who could correctly recognize convolutional neural networks rose from 21% to 81%. Participants’ confidence in their understanding of AI rose by more than 1 full point on a 5-point Likert scale (2.2 to 3.5, $p<.001$), while familiarity with AI ethics similarly improved (3.2 to 4.1, $p<.001$) (see Table 1).

Attitudes regarding AI, assessed after the workshops, were remarkable. 93.8% of participants believed that AI would significantly impact the future of healthcare. 88.9% of participants believed that it is important for future healthcare providers to learn about AI at a foundational level. 70.37% of participants agreed or strongly agreed that they would find it useful for their professional program to have more formal teaching on AI.

Collectively, these findings demonstrate that medical learners currently have relatively low baseline knowledge regarding AI. However, the increases in both knowledge and self-reported confidence in AI and its appraisal demonstrate that student-led workshops are effective and viable strategies for enhancing AI literacy among future clinicians. Importantly, the positive reception and enthusiasm for AI expressed by participants highlights the demand for the integration of structured AI education into the curriculum. While short extracurricular workshops cannot substitute for comprehensive training on AI, they can serve as effective primers and pique student interest.

Given the rapid evolution and integration of AI into clinical practice, today's medical trainees will become providers in an AI-native healthcare system, necessitating the integration of AI education into clinical curricula. Our experience suggests that peer-led educational models can help bridge the current lack of formal teaching while faculty-led curricular integration continues to evolve. We propose that student-led initiatives, supported by faculty, are a vital contribution that can complement broader efforts towards AI integration, while promoting diverse student perspectives. Next steps include wider dissemination across medical schools as well as advocacy efforts to support the formal integration of such workshops into medical curricula.

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Table 1

Pre- and post-intervention performance on multiple-choice knowledge questions, and pre- and post-intervention Likert scale questions. For multiple-choice knowledge questions, correct response rates are reported with McNemar test p-values, matched odds ratios (ORs) with 95% confidence intervals (CIs), and Holm-adjusted p-values. For Likert scale questions, mean (SD), median [IQR], Wilcoxon signed-rank test results, and Holm-adjusted p-values are shown.

MCQ Item	Answer	Pre Correct (%)	Post Correct (%)	Matched OR (95% CI)	Adjusted p-value
Which of the following best describes the term "machine learning"?	A type of artificial intelligence where computers identify patterns and make predictions without explicit programming	54.3	58.0	0.81 (0.39-1.69)	.710
Which type of machine learning involves training a model using labeled datasets, where the correct output is already known?	Supervised learning	24.7	90.1	0.02 (0.00-0.13)	<.001 (***)
In the context of AI, which of the following algorithms is most commonly used for image recognition tasks, such as identifying tumors in medical imaging?	Convolutional Neural Networks (CNNs)	21.2	81.2	0.04 (0.01-0.16)	<.001 (***)
Item	Pre Mean (SD)	Post Mean (SD)	Δ Mean	Wilcoxon W	Adjusted p-value
Items compared pre- and post-intervention					
As a future healthcare provider, I believe that it is important to learn about AI at a foundational level.	4.42 (0.72)	4.41 (0.69)	-0.01	145.0	.876

I feel confident in my understanding of artificial intelligence.	2.19 (0.79)	3.49 (0.63)	+1.31	0.0	<.001 (***)
I am familiar with the ethical concerns related to using AI in healthcare decision-making.	3.17 (0.86)	4.09 (0.60)	+0.91	61.5	<.001 (***)
I believe AI will significantly impact the future of healthcare.	4.33 (0.74)	4.49 (0.61)	+0.16	161.0	.136
Items only measured once					
I would be comfortable appraising the quality of data outputs generated by AI algorithms in a healthcare context.	2.49 (0.94)	-	-	-	-
This event was valuable for me as a future healthcare provider.	-	4.32 (0.73) Satisfied students (4/5): 86.42%	-	-	-
I would find it useful if my program had more formal teaching on AI.	-	4.06 (0.87) Satisfied students (4/5): 70.37%	-	-	-